

Online JEPA Composite Stability

A Two-Time-Scale Extension of Borkar (2008)

for Recursive Controlled Systems with Learned Substrates

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Abstract

Scaffold placeholder. Paper 0 of the Recursive Controlled Systems series proves a per-level stability budget $\lambda_k = \gamma_k - L_\theta \rho_k - L_d \eta_k - \beta \bar{\tau}_k - \frac{\ln \nu}{\tau_a}_k$ under hand-set Lyapunov functions and hand-set constants. This paper generalises the budget to the setting in which the substrate V_k itself is *learned online* by a Joint-Embedding Predictive Architecture (JEPA), yielding two new cost terms: an *online encoder cost* $L_{o\theta}(\Delta, \mu)$ derived by extending Borkar’s Ch. 6 two-time-scale stochastic approximation to discrete deployment events, and a *head cost* $L_H \kappa_k$ that bounds substrate sensitivity to head-specific payload variations. The new bound reduces to Paper 0’s Theorem VI exactly in the Paper 0 regime ($\Delta \rightarrow \infty, \kappa_k = 0$) — the encoder cost vanishes asymptotically and the head cost is identically zero when no L0 head is engaged at level k . Under an engaged head with $L_H \leq 1$ the head cost is bounded by κ_k but does not vanish; the bound is then strictly tighter than P0 and gates a CI-enforced production rollout via the empirical-constants protocol of Section 6. The final abstract will be written in T5 once the constants table from Q3 substrate measurements is in hand. See `PROTOCOL.md` for the locked theorem statement and proof skeleton.

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- 2 Related Work**
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References

- [1] Christopher M. Bishop. *Pattern Recognition and Machine Learning*. Information Science and Statistics. Springer, 2006. ISBN 978-0-387-31073-2. Ch. 10 (variational inference) is the standard reference for the KL-bound lemma supporting Paper 5 (H8d).
- [2] Vivek S. Borkar. *Stochastic Approximation: A Dynamical Systems Viewpoint*. Cambridge University Press / Hindustan Book Agency, 2008. ISBN 978-0-521-51592-4. Ch. 6 is the foundational two-time-scale stochastic approximation reference; extended by Paper 5 T1 (BRO-989) to discrete deployment events.
- [3] John N. Tsitsiklis and Benjamin Van Roy. An analysis of temporal-difference learning with function approximation. *IEEE Transactions on Automatic Control*, 42(5):674–690, 1997. doi: 10.1109/9.580874.